

SALT Device Models

1 Scope and conventions

SALT solves the periodic steady state of a balanced, single-harmonic transmission network using EMT device models, with the system frequency ω_s as an unknown. EMT integrates the same device models in the time domain with the trapezoidal rule.

Every device contributes a residual $F(x) = 0$. Newton's method is written in the stamped form

$$J(x^{[k]}) x^{[k+1]} = J(x^{[k]}) x^{[k]} - F(x^{[k]}), \quad (1)$$

where $x^{[k]}$ is the vector of unknowns at iteration k , F is the stacked vector of device residuals and $J = \partial F / \partial x$ is the system Jacobian. Written this way, a linear equation contributes its coefficients to J and its constant to the right-hand side directly.

In the steady-state formulation the network variables are phasors carried as real and imaginary components $[X_{re}, X_{im}]$ at the fundamental, and machine states in their own rotating frame are DC. Consequently every d/dt acting on a rotating frame state vanishes, which is noted equation by equation below.

1.1 Per-unit bases

Quantity	Base	Convention
Bus voltage	$V_{\text{base}}^{\text{bus}}$	line-to-line, RMS, kV
Bus current	$S_{\text{base}} / (V_{\text{base}}^{\text{bus}} \sqrt{3})$	kA
Machine voltage	$V_{\text{base}}^{\text{gen}} = V_{\text{base}}^{\text{bus}} \cdot 10^3 \sqrt{2/3}$	line-to-neutral, peak, V
Machine current	$I_{\text{base}}^{\text{gen}} = S_{\text{base}}^{\text{gen}} / (V_{\text{base}}^{\text{gen}} \cdot 3/2)$	peak, A
Inverter	$V_{\text{base}}^{\text{ibr}}, I_{\text{base}}^{\text{ibr}} = S^{\text{ibr}} / (V_{\text{base}}^{\text{ibr}} \sqrt{3})$	line-to-line, RMS
Induction motor	as machine	line-to-neutral, peak

Because $V_{\text{base}}^{\text{gen}} = V_{\text{base}}^{\text{bus}} \sqrt{2/3}$ and the conversion from line-to-neutral peak to line-to-line RMS carries $\sqrt{3/2}$, the machine-to-bus *voltage* scale factor is exactly 1. The remaining scale factors $c_I^{\text{gen} \rightarrow \text{bus}}$ and $c_V^{\text{ibr} \rightarrow \text{bus}}$ are:

$$c_I^{\text{gen} \rightarrow \text{bus}} = \frac{I_{\text{base}}^{\text{gen}} / \sqrt{2}}{10^3 I_{\text{base}}^{\text{bus}}}, \quad c_V^{\text{ibr} \rightarrow \text{bus}} = \frac{V_{\text{base}}^{\text{ibr}}}{V_{\text{base}}^{\text{bus}}}, \quad (2)$$

where $c_I^{\text{gen} \rightarrow \text{bus}}$ converts machine-base current to bus-base current, $c_V^{\text{ibr} \rightarrow \text{bus}}$ converts inverter-base voltage to bus-base voltage, $I_{\text{base}}^{\text{gen}}$ and $I_{\text{base}}^{\text{bus}}$ are the machine and bus current bases, and $V_{\text{base}}^{\text{ibr}}$ and $V_{\text{base}}^{\text{bus}}$ are the inverter and bus voltage bases. Controller parameters (exciter, governor) are per unit on the machine base and are used without conversion.

1.2 Sign convention

Machines and inverters produce power, so their terminal current leaves the device and enters the network; it is stamped negative into the bus equation. Loads consume power and are stamped positive.

1.3 dq -Transform

In SALT, the three-phase frequency-domain grid quantities interface with the dq -quantities through a phase shift,¹ circumventing explicit computation of the dq -transform. Different device models use different conventions of the dq -transform, which are accommodated via linear transformations. The equations for each device are as follows. Throughout, θ is the angle of the device's own rotating frame, $\theta_a = \theta$, $\theta_b = \theta - \frac{2\pi}{3}$ and $\theta_c = \theta + \frac{2\pi}{3}$ are the three phase angles, and c_V , c_I are the voltage and current base-conversion factors of §1.1.

Synchronous machine and induction motor.

The dq -transform and its inverse are

$$\begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta_a & \cos \theta_b & \cos \theta_c \\ -\sin \theta_a & -\sin \theta_b & -\sin \theta_c \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}, \quad \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} \cos \theta_a & -\sin \theta_a & 1 \\ \cos \theta_b & -\sin \theta_b & 1 \\ \cos \theta_c & -\sin \theta_c & 1 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}, \quad (3)$$

where $i_{a,b,c}$ are the instantaneous phase currents, i_d, i_q are their direct- and quadrature-axis components in the rotating frame, and i_0 is the zero-sequence component, which vanishes under the balanced assumption. The SALT equivalent operates on the phase- a phasor:

$$\begin{bmatrix} I_{\phi a, re} \\ I_{\phi a, im} \end{bmatrix} = c_I T_{\text{gen}}^{-1}(\theta) \begin{bmatrix} i_d \\ i_q \end{bmatrix}, \quad T_{\text{gen}}^{-1}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad (4)$$

where $I_{\phi a, re}$ and $I_{\phi a, im}$ are the real and imaginary components of the phase- a current phasor.

Grid-forming inverter.

The inverter convention negates the quadrature axis,

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta_a & \cos \theta_b & \cos \theta_c \\ \sin \theta_a & \sin \theta_b & \sin \theta_c \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}, \quad (5)$$

so the SALT equivalent is the machine phase shift composed with a sign flip on q :

$$\begin{bmatrix} I_{\phi a, re} \\ I_{\phi a, im} \end{bmatrix} = c_I T_{\text{ibr}}^{-1}(\theta) \begin{bmatrix} i_{o,d} \\ i_{o,q} \end{bmatrix}, \quad T_{\text{ibr}}^{-1}(\theta) = T_{\text{gen}}^{-1}(\theta) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}, \quad (6)$$

where $i_{o,d}, i_{o,q}$ are the inverter output-current components.

Composite load.

The load frame is the system reference, $\theta_{\text{sys}} = 0$, so the phase shift is the identity and the transform disappears entirely:

$$\begin{bmatrix} v_d^s \\ v_q^s \end{bmatrix} = c_V \begin{bmatrix} V_{\phi a, re} \\ V_{\phi a, im} \end{bmatrix}, \quad (7)$$

where v_d^s, v_q^s are the stator voltage components and $V_{\phi a, re}, V_{\phi a, im}$ the real and imaginary components of the phase- a bus voltage phasor.

¹Because SALT writes complex quantities as real and imaginary components, the phase shift $Y = e^{j\theta} X$ is computed as $\begin{bmatrix} Y_{re} \\ Y_{im} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} X_{re} \\ X_{im} \end{bmatrix}$.

Balanced three-phase output.

The remaining two phases are enforced algebraically rather than solved: for a phase-*a* phasor I_a ,

$$\begin{bmatrix} I_b \\ I_c \end{bmatrix} = \begin{bmatrix} e^{-j2\pi/3} \\ e^{+j2\pi/3} \end{bmatrix} I_a, \quad (8)$$

which in real and imaginary components reads

$$\begin{aligned} I_{b,re} &= -\frac{1}{2}I_{a,re} + \frac{\sqrt{3}}{2}I_{a,im}, & I_{b,im} &= -\frac{\sqrt{3}}{2}I_{a,re} - \frac{1}{2}I_{a,im}, \\ I_{c,re} &= -\frac{1}{2}I_{a,re} - \frac{\sqrt{3}}{2}I_{a,im}, & I_{c,im} &= \frac{\sqrt{3}}{2}I_{a,re} - \frac{1}{2}I_{a,im}, \end{aligned} \quad (9)$$

where I_b and I_c are the phase-*b* and phase-*c* phasors and the subscripts *re*, *im* denote their real and imaginary components.

1.4 Document convention

Each device model below is stated twice. The first form is the governing time-domain equation, written with its derivatives intact, as EMT integrates it. An arrow then points to the steady-state form that SALT stamps:

$$\text{time-domain equation} \xrightarrow{\text{SS}} \text{steady-state residual stamped by SALT.}$$

Reading left to right therefore shows exactly which terms survive the steady-state assumption and which vanish. Equations given in one form only are already identical in both solvers.

2 Synchronous machine: GENROU²

Round-rotor machine with one field winding (*fd*), one *d*-axis damper (*1d*) and two *q*-axis dampers (*1q*, *2q*).

2.1 Circuit matrices from nameplate data

The winding inductances and rotor resistances of the equivalent circuit are not given directly; they are computed from the nameplate reactances and open-circuit time constants.

$$L_{ad} = X_d - X_l, \quad L_{aq} = X_q - X_l, \quad L_\ell = X_l, \quad L_0 = X_l, \quad (10)$$

$$L_{ffd} = \frac{L_{ad}^2}{X_d - X_d'}, \quad L_{f1d} = L_{ad}, \quad L_{11d} = \frac{L_{ad}^2}{X_d - X_d'}, \quad (11)$$

$$L_{11q} = \frac{L_{aq}^2}{X_q - X_q'}, \quad L_{22q} = \frac{L_{aq}^2}{X_q - X_q''}. \quad (12)$$

where X_d, X_q are the synchronous reactances, X_d', X_q' the transient reactances, X_d'' the subtransient reactance, X_l the leakage reactance, L_{ad}, L_{aq} the *d*- and *q*-axis mutual inductances, L_ℓ the leakage inductance, L_0 the zero-sequence inductance, L_{ffd} the field self-inductance, L_{f1d} the field-to-damper mutual inductance, L_{11d} the *d*-axis damper self-inductance, and L_{11q}, L_{22q} the *q*-axis damper self-inductances.

²P. Kundur, *Power System Stability and Control*: stator equations 3.120–3.134, electrical torque 3.174, machine parameter derivation §4.1, per-unit bases §3.4.1 and §3.4.9, Park transform 3.60, initialisation §3.6.2, current-direction conventions Fig. 3.9 (generator) and Figs. 7.6–7.7 (load).

Rotor resistances follow from the open-circuit time constants. With

$$L_{fd} = \frac{(X'_d - X_l)L_{ad}}{L_{ad} - (X'_d - X_l)}, \quad z = X''_d - X_l, \quad y = \frac{L_{ad}L_{fd}}{L_{ad} + L_{fd}}, \quad L_{1d} = \frac{yz}{y - z}, \quad (13)$$

$$R_{1d} = \frac{y + L_{1d}}{T''_{d0}}, \quad R_{fd} = \frac{L_{ad} + L_{fd}}{T'_{d0}}, \quad (14)$$

and analogously on the q axis with $L_{1q} = (X'_q - X_l)L_{aq}/(L_{aq} - (X'_q - X_l))$, $y = L_{aq}L_{1q}/(L_{aq} + L_{1q})$, $L_{2q} = yz/(y - z)$:

$$R_{2q} = \frac{y + L_{2q}}{T''_{q0}}, \quad R_{1q} = \frac{L_{aq} + L_{1q}}{T'_{q0}}. \quad (15)$$

where T'_{d0}, T'_{q0} are the open-circuit transient time constants, T''_{d0}, T''_{q0} the open-circuit subtransient time constants, R_{fd} the field resistance, R_{1d} the d -axis damper resistance, R_{1q}, R_{2q} the q -axis damper resistances, $L_{fd}, L_{1d}, L_{1q}, L_{2q}$ the corresponding winding leakage inductances, and y, z intermediate quantities carrying no physical meaning on their own.

The assembled matrices are

$$\mathbf{R} = \begin{bmatrix} R_a & & & & & & \\ & R_a & & & & & \\ & & R_a & & & & \\ & & & -R_{fd} & & & \\ & & & & -R_{1d} & & \\ & & & & & -R_{1q} & \\ & & & & & & -R_{2q} \end{bmatrix}, \quad (16)$$

where R_a is the armature (stator) resistance, the rotor entries carrying a negative sign from the equivalent-circuit orientation, and

$$\mathbf{L} = \begin{bmatrix} -(L_{ad} + L_\ell) & & & L_{ad} & L_{ad} & & \\ & -(L_{aq} + L_\ell) & & & & L_{aq} & L_{aq} \\ & & -L_0 & & & & \\ -L_{ad} & & & L_{ffd} & L_{f1d} & & \\ -L_{ad} & & & L_{f1d} & L_{11d} & & \\ & -L_{aq} & & & & L_{11q} & L_{aq} \\ & -L_{aq} & & & & L_{aq} & L_{22q} \end{bmatrix}. \quad (17)$$

$\mathbf{L}_{sp}$ holds the flux terms that multiply rotor speed: it is $\mathbf{L}$ with the d and q rows exchanged and signed,

$$\mathbf{L}_{sp} = \begin{bmatrix} & (L_{aq} + L_\ell) & & & & -L_{aq} & -L_{aq} \\ -(L_{ad} + L_\ell) & & & L_{ad} & L_{ad} & & \\ & & & & & & \end{bmatrix}. \quad (18)$$

2.2 Equations

Let $i = [i_d, i_q, i_{fd}]^\top$, $e = [e_d, e_q, e_{fd}]^\top$, and $\mathbf{R}, \mathbf{L}_{sp}$ denote the (d, q, fd) sub-blocks.

Stator.

$$e = -\mathbf{R}i + \mathbf{L} \frac{di}{dt} + \mathbf{L}_{sp} i \frac{\omega_r}{\omega_0} \xrightarrow{\text{SS}} 0 = -e + \left(-\mathbf{R} + \mathbf{L}_{sp} \frac{\omega_r}{\omega_0} \right) i. \quad (19)$$

where i_d, i_q are the stator currents in the rotor frame, i_{fd} the field current, e_d, e_q the corresponding stator voltages, e_{fd} the field voltage, ω_r the rotor electrical speed and ω_0 the nominal speed.

dq-transform, voltage.

$$0 = -v_{\phi a} + T_{\text{gen}}^{-1}(\theta) c_V e_{dq}. \quad (20)$$

where $v_{\phi a}$ is the phase- a terminal voltage phasor and $T_{\text{gen}}^{-1}(\theta)$ is the phase shift of Equation (4).

Field voltage.

Supplied by the exciter.

Rotor speed.

$$\omega_r = \omega_s. \quad (21)$$

where ω_s is the system frequency. In steady state every machine turns at the system frequency, so the rotor speed is not an independent unknown.

Swing.

Electrical torque is derived from the speed terms of the stator equation, so when $\omega_r \neq \omega_0$ the mechanical torque is normalised:

$$T_m \frac{\omega_r}{\omega_0} = e_d i_d + e_q i_q + R_a (i_d^2 + i_q^2). \quad (22)$$

where T_m is the mechanical torque supplied by the governor.

In steady state the rotor acceleration is zero, so the swing equation $\frac{2H}{\omega_0} \frac{d\omega_r}{dt} = T_m - T_e$ collapses to $T_m = T_e$ and is replaced by the conservation of power expressed in Equation (22): the mechanical power entering the shaft, $T_m \omega_r / \omega_0$, equals the electrical power leaving the machine. The first two terms on the right-hand side are the electrical power delivered at the terminals, and the armature term $R_a (i_d^2 + i_q^2)$ is the copper loss dissipated in transferring it.

dq-transform, current.

$$0 = -I_{\phi a} + T_{\text{gen}}^{-1}(\theta) c_I i_{dq}. \quad (23)$$

where $I_{\phi a}$ is the phase- a terminal current phasor.

Balanced output current.

Equation (8).

Mechanical torque.

Supplied by the governor.

3 Exciter: EXDC1

Five controller states $x_1 \dots x_5$ plus an algebraic terminal-voltage measurement v_t .

$$0 = -K_e x_1 - T_e \dot{x}_1 + x_4 \xrightarrow{\text{SS}} 0 = -K_e x_1 + x_4, \quad (24)$$

$$0 = x_2 + T_r \dot{x}_2 - v_t \xrightarrow{\text{SS}} 0 = x_2 - v_t, \quad (25)$$

$$0 = x_2 + T_c \dot{x}_2 + x_3 + T_b \dot{x}_3 + x_5 + T_c \dot{x}_5 - v_{\text{ref}} \xrightarrow{\text{SS}} 0 = x_2 + x_3 + x_5 - v_{\text{ref}}, \quad (26)$$

$$0 = -K_a x_3 + x_4 + T_a \dot{x}_4 \xrightarrow{\text{SS}} 0 = -K_a x_3 + x_4, \quad (27)$$

$$0 = -K_{f1} \dot{x}_1 + x_5 + T_{f1} \dot{x}_5 \xrightarrow{\text{SS}} 0 = x_5. \quad (28)$$

where x_1 is the exciter output voltage, x_2 the measured terminal voltage, x_3 the lead-lag compensator output, x_4 the regulator output, x_5 the rate-feedback state, v_{ref} the voltage setpoint, K_e and T_e the exciter gain and time constant, T_r the measurement time constant, T_b and T_c the lead-lag time constants, K_a and T_a the regulator gain and time constant, and K_{f1} and T_{f1} the rate-feedback gain and time constant.

Terminal voltage, in machine per unit:

$$0 = -v_t + \sqrt{\frac{2}{3}} \left(v_a^2 + v_b^2 + v_c^2 \right)^{1/2}. \quad (29)$$

where v_t is the terminal-voltage magnitude and v_a, v_b, v_c are the instantaneous phase voltages.

The exciter drives the machine field-voltage equation through

$$0 = c_{g \rightarrow e} e_{fd} - x_1 \frac{\omega_r}{\omega_0}, \quad c_{g \rightarrow e} = \frac{L_{ad}}{R_{fd}}. \quad (30)$$

where $c_{g \rightarrow e}$ converts the machine field voltage to the exciter per-unit base.

4 Governor: TGOV1D³

Two states plus an algebraic mechanical power.

$$0 = x_1 + T_3 \dot{x}_1 - x_2 - T_2 \dot{x}_2 \xrightarrow{\text{SS}} 0 = x_1 - x_2, \quad (31)$$

$$0 = p_{\text{ref}} - \left(\frac{\omega_r}{\omega_0} - 1 \right) - R x_2 - R T_1 \dot{x}_2 \xrightarrow{\text{SS}} 0 = p_{\text{ref}} - \left(\frac{\omega_r}{\omega_0} - 1 \right) - R x_2, \quad (32)$$

$$0 = x_1 - \left(\frac{\omega_r}{\omega_0} - 1 \right) D_t - p_{\text{mech}}. \quad (33)$$

where x_1 is the turbine power state, x_2 the valve position, p_{ref} the load reference setpoint, p_{mech} the mechanical power delivered to the shaft, R the droop, T_1 the governor time constant, T_2 and T_3 the turbine lead and lag time constants, and D_t the turbine damping coefficient.

Electrical and mechanical angles satisfy $\theta = (p/2)\theta_M$, hence $\omega_r = (p/2)\omega_M$, where p is the number of poles, θ_M the mechanical rotor angle and ω_M the mechanical speed. With $p = 2$ this gives $\omega_r = \omega_M$ and the mechanical torque coupling into the machine is

$$0 = p_{\text{mech}} - \frac{\omega_r}{\omega_0} T_m. \quad (34)$$

where T_m is the mechanical torque appearing in the machine swing equation.

³Governor per-unit conventions follow the IEEE PES report, https://site.ieee.org/fw-pes/files/2013/01/PES_TR1.pdf.

5 Grid-forming inverter⁴

Droop-controlled inverter behind an LCL filter, with an inner current loop, an outer voltage loop and a virtual resistor. All dq quantities are two-vectors; $\mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ represents multiplication by j .

Network interface.

$$0 = -I_{\phi a} + T_{\text{ibr}}^{-1}(\theta) c_I i_{o,dq}, \quad (35)$$

plus Equation (8), and

$$0 = -v_{\phi a} + T_{\text{ibr}}^{-1}(\theta) c_V v_{g,dq}. \quad (36)$$

where $i_{o,dq}$ is the output current at the grid side of the filter, $v_{g,dq}$ the grid-side voltage, and $T_{\text{ibr}}^{-1}(\theta)$ the phase shift of Equation (6).

Passive network.

$$\dot{i}_o = \frac{1}{L_c} (-r_c i_o + v_o - v_g) + j\omega i_o \xrightarrow{\text{SS}} 0 = \frac{1}{L_c} (-r_c i_o + v_o - v_g) + \frac{\omega_r}{\omega_0} \mathbf{J} i_o, \quad (37)$$

$$\dot{v}_o = \frac{1}{C_f} (i_f - i_o) + j\omega v_o \xrightarrow{\text{SS}} 0 = \frac{1}{C_f} (i_f - i_o) + \frac{\omega_r}{\omega_0} \mathbf{J} v_o, \quad (38)$$

$$\dot{i}_f = \frac{1}{L_f} (-r_f i_f + v_i - v_o) + j\omega i_f \xrightarrow{\text{SS}} 0 = \frac{1}{L_f} (-r_f i_f + v_i - v_o) + \frac{\omega_r}{\omega_0} \mathbf{J} i_f. \quad (39)$$

where L_c and r_c are the grid-side inductance and resistance, L_f and r_f the converter-side inductance and resistance, C_f the filter capacitance, i_f the converter-side current, v_o the filter-capacitor voltage and v_i the converter terminal voltage.

Current controller.

$$v_i = k_{C,i} \gamma + k_{C,p} \dot{\gamma} - j\omega_r L_f i_f + G_C v_o \xrightarrow{\text{SS}} v_i = k_{C,i} \gamma - \frac{\omega_r}{\omega_0} L_f \mathbf{J} i_f + G_C v_o, \quad (40)$$

$$\dot{\gamma} = i_f^{\text{ref}} - i_f \xrightarrow{\text{SS}} 0 = i_f^{\text{ref}} - i_f. \quad (41)$$

where γ is the current-loop integrator state, i_f^{ref} the current reference from the voltage loop, $k_{C,p}$ and $k_{C,i}$ the proportional and integral gains, and G_C the voltage feed-forward gain.

Voltage controller.

$$i_f^{\text{ref}} = k_{V,i} x + k_{V,p} \dot{x} - j\omega_r C_f v_o + G_V i_o \xrightarrow{\text{SS}} i_f^{\text{ref}} = k_{V,i} x - \frac{\omega_r}{\omega_0} C_f \mathbf{J} v_o + G_V i_o, \quad (42)$$

$$\dot{x} = v_o^{\text{ref}} - v_o \xrightarrow{\text{SS}} 0 = v_o^{\text{ref}} - v_o. \quad (43)$$

where x is the voltage-loop integrator state, v_o^{ref} the capacitor-voltage reference, $k_{V,p}$ and $k_{V,i}$ the proportional and integral gains, and G_V the current feed-forward gain.

⁴B. Kenyon et al., “Open-Source PSCAD Grid-Following and Grid-Forming Inverters and a Benchmark for Zero-Inertia Power System Simulations” (NREL), which is also the source of the default parameter set; and “An Accurate Small-Signal Model of Inverter-Dominated Islanded Microgrids using dq Reference Frame”.

Droop.

With $M_p = \text{droop}_P \omega_0$ and $M_q = \text{droop}_Q$,

$$\omega_r = \omega_0 + M_p(p_{\text{set}} - p_{\text{avg}}), \quad (44)$$

$$v_{o,q}^{\text{ref,droop}} = v_{o,q}^{\text{set}} + M_q(q_{\text{set}} - q_{\text{avg}}), \quad (45)$$

$$\dot{p}_{\text{avg}} = \omega_{\text{meas}}(p - p_{\text{avg}}), \quad p = v_{g,dq}^\top i_{o,dq} \xrightarrow{\text{SS}} 0 = p - p_{\text{avg}}, \quad (46)$$

$$\dot{q}_{\text{avg}} = \omega_{\text{meas}}(q - q_{\text{avg}}), \quad q = v_{g,dq}^\top \mathbf{J} i_{o,dq} \xrightarrow{\text{SS}} 0 = q - q_{\text{avg}}, \quad (47)$$

$$\dot{\theta} = \omega_r \xrightarrow{\text{SS}} \omega_s = \omega_r. \quad (48)$$

where M_p and M_q are the active- and reactive-power droop gains, p_{set} and q_{set} the power setpoints, p and q the instantaneous active and reactive power at the grid side, p_{avg} and q_{avg} their low-pass filtered values, ω_{meas} the measurement filter cutoff, $v_{o,q}^{\text{set}}$ the nominal quadrature voltage setpoint and $v_{o,q}^{\text{ref,droop}}$ the droop-adjusted reference.

No factor of 3/2 appears in p and q . It would arise from the dq -transform for peak, line-to-neutral, non-per-unit quantities; with RMS, line-to-line, per-unit quantities it cancels.

Virtual resistor.

With $\omega_{VR} = \omega_{VR}^{\text{rad}}/(2\pi)$,

$$\frac{d}{dt} V_o^{VR,\text{avg}} = \omega_{VR}(R_v i_o - V_o^{VR,\text{avg}}) \xrightarrow{\text{SS}} 0 = \omega_{VR}(R_v i_o - V_o^{VR,\text{avg}}), \quad (49)$$

$$V_o^{VR} = V_o^{VR,\text{avg}} - R_v i_o, \quad (50)$$

$$v_o^{\text{ref}} = [0, v_{o,q}^{\text{ref,droop}}]^\top + V_o^{VR}. \quad (51)$$

where R_v is the virtual resistance, ω_{VR} the virtual-resistor filter cutoff, $V_o^{VR,\text{avg}}$ the filtered virtual-resistor voltage and V_o^{VR} the voltage correction it applies to the reference.

6 Composite load: induction motor + ZIP⁵

One induction motor in parallel with one static ZIP load at each load bus. The motor share of the bus MW is `powerPercent`.

6.1 Induction motor

Let $i_{sr} = [i_d^s, i_q^s, i_d^r, i_q^r]^\top$ and $v_{sr} = [v_d^s, v_q^s, 0, 0]^\top$.

⁵EPRI, “Technical Reference on the Composite Load Model” (2020), shows that the motor’s reactive power cannot be specified by a power factor and must come from the slip. As such, initialisation requires its internal Newton–Raphson solve. The default parameter set comes from the following. The $Z/I/P$ split and the motor share are from “Implementation of the WECC Composite Load Model for Utilities using the Component-Based Modeling Approach”, <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7520081>. Reactive power is taken as pure impedance for industrial loads, following NERC, “Reliability Guideline: Developing Load Model Composition Data” (March 2017), https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/lmwg/reliability_guideline_-_load_model_composition_-_2017-02-28.pdf. The ZIP low-pass filter time constant is the Simscape default, <https://www.mathworks.com/help/sps/ref/constantpowerloadthreephase.html>. Motor parameters are Kundur’s small industrial motor, §7.4.3.c; the motor transform is Kundur 7.23–7.27 and the slip/efficiency relation §7.7.2. Supporting literature for the simplified IM+ZIP composition comes from “Development of Composite Load Models of Power Systems using On-line Measurement Data” (2006), “Standard Load Models for Power Flow and Dynamic Performance Simulation” (1995), and “Load Modeling — A Review” (2018).

Terminal interface.

$$0 = -v_{dq}^s + c_V v_{\phi a}, \quad 0 = -(i_{dq}^s + i_{dq}^{\text{ZIP}}) + c_I I_{\phi a}, \quad (52)$$

plus Equation (8), where v_{dq}^s and i_{dq}^s are the motor stator voltage and current and i_{dq}^{ZIP} is the current drawn by the static ZIP load sharing the bus.

Slip and reference angle.

$$\omega_{\text{slip}} = \omega_s - \omega_r, \quad \theta_{\text{sys}} = 0. \quad (53)$$

where ω_{slip} is the slip frequency, ω_r the rotor speed and θ_{sys} the system reference angle, fixed at zero.

Stator and rotor.

With

$$\mathbf{R} = \text{diag}(R_s, R_s, R_r, R_r), \quad \mathbf{L} = \begin{bmatrix} L_{ss}\mathbf{I}_2 & L_m\mathbf{I}_2 \\ L_m\mathbf{I}_2 & L_{rr}\mathbf{I}_2 \end{bmatrix}, \quad (54)$$

$$\mathbf{\Omega} = \begin{bmatrix} 0 & -\omega_s & 0 & 0 \\ \omega_s & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega_{\text{slip}} \\ 0 & 0 & \omega_{\text{slip}} & 0 \end{bmatrix}, \quad (55)$$

$$v_{sr} = \mathbf{R} i_{sr} + \frac{d}{dt} \frac{\mathbf{L}}{\omega_0} i_{sr} + \frac{\mathbf{\Omega}}{\omega_0} \mathbf{L} i_{sr} \xrightarrow{\text{SS}} 0 = -v_{sr} + \mathbf{R} i_{sr} + \mathbf{\Omega} \frac{\mathbf{L}}{\omega_0} i_{sr}. \quad (56)$$

where R_s and R_r are the stator and rotor resistances, L_{ss} , L_{rr} and L_m the stator, rotor and magnetising inductances, $\mathbf{\Omega}$ the speed-coupling matrix and i_{sr} , v_{sr} the stacked stator and rotor currents and voltages.

The inductances follow from the nameplate reactances as $L_m = X_m$, $L_{ss} = X_s + X_m$, $L_{rr} = X_r + X_m$, where X_m is the magnetising reactance and X_s , X_r the stator and rotor leakage reactances; no division by ω is needed because the reactances are per unit at $\omega_{pu} = 1$.

Swing.

$$\text{With } \mathbf{L}_{Te} = \mathbf{L}^\top \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\frac{2H}{\omega_0} \frac{d\omega_r}{dt} = i_{sr}^\top \mathbf{L}_{Te} i_{sr} - T_0 \left(\frac{\omega_r}{\omega_s} \right)^m \xrightarrow{\text{SS}} 0 = i_{sr}^\top \mathbf{L}_{Te} i_{sr} - T_0 \left(\frac{\omega_r}{\omega_s} \right)^m. \quad (57)$$

where H is the motor inertia constant, $\mathbf{L}_{Te}$ the matrix that extracts electrical torque from the currents, T_0 the load torque at synchronous speed and m the torque-speed exponent of the mechanical load. As for the synchronous machine, the steady-state form states that the electrical torque produced balances the mechanical torque absorbed.

Base.

The motor MVA base is its apparent power at full loading,

$$S_{\text{base}}^{\text{IM}} = \frac{P_{\text{bus}} \cdot \text{powerPercent}}{\text{PF}}, \quad (58)$$

where P_{bus} is the total active power of the bus, **powerPercent** the motor's share of it and PF the assumed power factor. Voltage and current bases are derived exactly as for the synchronous machine. PF only sizes the base; the motor's reactive power is determined by the slip, not by PF.

6.2 ZIP

$$\begin{bmatrix} P \\ Q \end{bmatrix} = \begin{bmatrix} P_z \\ Q_z \end{bmatrix} V_m^2 + \begin{bmatrix} P_i \\ Q_i \end{bmatrix} V_m + \begin{bmatrix} P_p \\ Q_p \end{bmatrix}, \quad (59)$$

$$V_m = \sqrt{(v_d^s)^2 + (v_q^s)^2}, \quad (60)$$

$$i_d^{\text{ZIP}} = \frac{Pv_d + Qv_q}{V_m^2}, \quad i_q^{\text{ZIP}} = \frac{Pv_q - Qv_d}{V_m^2}. \quad (61)$$

where P and Q are the active and reactive power drawn by the static load, V_m the terminal-voltage magnitude, P_z, Q_z the constant-impedance shares, P_i, Q_i the constant-current shares, P_p, Q_p the constant-power shares and $i_d^{\text{ZIP}}, i_q^{\text{ZIP}}$ the resulting load currents.

The Z, I, P shares are normalised within the ZIP category, so they sum to one for P and independently for Q .

7 Network elements

7.1 Transmission line

Three-phase nominal- π with no mutual coupling. From the case data, $L = X/\omega_0$ and $C = B_{\text{chg}}/(2\omega_0)$, where X is the branch reactance and B_{chg} the total line-charging susceptance.

$$Y_{br} = \frac{1}{R + j\omega L}, \quad Y_{sh} = j\omega C, \quad (62)$$

$$I_{\text{from}} = Y_{br}(V_f - V_t) + Y_{sh}V_f, \quad I_{\text{to}} = -Y_{br}(V_f - V_t) + Y_{sh}V_t. \quad (63)$$

where Y_{br} is the series branch admittance, Y_{sh} the shunt admittance at each end, R the branch resistance, V_f and V_t the from- and to-bus voltage phasors, and $I_{\text{from}}, I_{\text{to}}$ the currents injected at each end.

Because ω is an unknown, each branch also contributes $\partial I/\partial\omega$ to the Jacobian column of the frequency variable:

$$\frac{\partial I_{br}}{\partial\omega} = -\left(\frac{jL}{|Z|^2} + \frac{2\omega L^2(R - j\omega L)}{|Z|^4}\right)(V_f - V_t), \quad \frac{\partial I_{sh}}{\partial\omega} = jC V. \quad (64)$$

where $Z = R + j\omega L$ is the series branch impedance and V is the terminal voltage at the end where the shunt is stamped.

7.2 Transformer

Series R - L with $L = X/\omega_0$. When $R \neq 0$ it is stamped as an admittance $Y = 1/(R + j\omega L)$. When $R = 0$ there is no admittance, so the branch current becomes an explicit unknown (modified nodal analysis) and the branch contributes

$$0 = -(V_f - V_t) + j\omega L I, \quad \frac{\partial}{\partial\omega} = jL I. \quad (65)$$

where I is the branch current carried as an explicit unknown and L the leakage inductance.

7.3 Constant-impedance load

Selected by the sign of the reactance:

$$Y = \begin{cases} \frac{1}{R + j\omega L}, & X > 0 \quad (L = X/\omega_0), \\ \frac{1}{R} + j\omega C, & X < 0 \quad (C = \text{Im}(1/Z)/\omega_0), \\ \frac{1}{R}, & X = 0. \end{cases} \quad (66)$$

where Y is the stamped shunt admittance, R and X the load resistance and reactance from the case data, $Z = R + jX$, and L , C the equivalent inductance or capacitance the reactance corresponds to.

7.4 Slack-angle equation

The system frequency adds one unknown, so one equation is added: the slack bus angle is pinned by fixing the ratio of the imaginary to the real part of its phase- a voltage, which is invariant to magnitude,

$$0 = c V_{\text{re}}^{\text{slack}} - V_{\text{im}}^{\text{slack}}, \quad c = \frac{\text{Im}(V_{\text{pf}}^{\text{slack}})}{\text{Re}(V_{\text{pf}}^{\text{slack}})}. \quad (67)$$

where $V_{\text{re}}^{\text{slack}}$ and $V_{\text{im}}^{\text{slack}}$ are the real and imaginary components of the slack-bus phase- a voltage, $V_{\text{pf}}^{\text{slack}}$ is the slack voltage from the power-flow solution and c is the tangent of the angle it fixes.

8 System frequency

The system frequency in SALT is a variable that is used by device models and network branch elements, as previously described. It is not explicitly defined by its own equation, instead emerging naturally from power mismatches, and equation (67) keeps the system full-rank. However, during transient simulation, power mismatches are not resolved explicitly by frequency deviation due to the dynamic flow of power, requiring EMT to explicitly define the system frequency.

EMT carries the system frequency as an MVA-weighted average of every rotating machine and inverter speed:

$$f_{s0} = \frac{\sum_i S_{\text{base},i} f_i}{\sum_i S_{\text{base},i}}. \quad (68)$$

where f_{s0} is the reported system frequency, $S_{\text{base},i}$ the MVA base of device i and f_i its instantaneous electrical frequency.